Adaptive Role Selection for Zero-Shot Reasoning

Rodrigo Pozo rfpozo@uc.cl
Cristóbal Vasquez cristobal.vasquez@uc.cl
Luis Miranda lmirandn@uc.cl
Oscar Loch oscar.loch@uc.cl

Abstract

LLMs have demonstrated significant progress in reasoning and comprehension tasks, yet their zero-shot reasoning abilities remain challenging. Prior work has shown that assigning specific roles to prompts enhances LLM performance, but manually creating these roles is labor-intensive. An innovative solution is proposed: using the LLM to automatically select the most appropriate roles for each task, optimizing time and resources. Through comparative experiments, it is demonstrated that automated role generation is both feasible and superior in effectiveness and adaptability. The approach is evaluated using datasets spanning arithmetic and commonsense reasoning. Results show that Role-Play Prompting is highly effective for the AQUA-RAT dataset, while Adaptive Role-Play surpasses all baselines, particularly in commonsense reasoning tasks. These findings suggest that LLMs can autonomously generate roles, improving zero-shot reasoning and reducing the need for manual intervention. This advancement enhances the scalability and adaptability of LLMs, providing new insights into self-optimization and adaptive learning in advanced language mode

1 Introduction

In the field of artificial intelligence and natural language processing, large language models (LLMs) have demonstrated remarkable progress in reasoning and comprehension tasks. However, the ability of these models to perform accurate reasoning in zero-shot contexts still presents significant challenges. Works such as "Better Zero-Shot Reasoning with Role-Play Prompting" (Kong et al., 2024) have shown that assigning specific roles to prompts can substantially improve the performance of LLMs in these tasks. Nevertheless, the manual creation of these roles is a laborious process. This project proposes an innovative solution: using the same LLM to automatically select the most suitable roles for each task, thus optimizing the time and resources required. Through experiments designed to compare this methodology with traditional approaches, the aim is to demonstrate that the automation of role generation is not only feasible but also superior in terms of efficiency and adaptability compared to other Zero-Shot Reasoning techniques.

This project aims to advance the automation of role generation for large language models (LLMs) to improve their performance in Zero-Shot Reasoning tasks. To implement these solutions, the following steps will be applied:

1. A specific instruction will be provided to the LLM to choose an appropriate assistant role to solve a given task.

2. Next, the LLM will use the generated role to provide a response to the original question.

The experiments designed to evaluate this methodology will compare the performance of the LLM with other Zero-Shot and Few-Shot Reasoning techniques present in the literature (Kojima et al., 2023), (Kong et al., 2024), (Wei et al., 2023), (Wu et al., 2023) along with the Zero-Shot baseline.

This approach has the potential to demonstrate that LLMs can autonomously generate roles, which will not only improve their zero-shot reasoning capability but also significantly reduce the need for manual intervention. This will facilitate the scalability and adaptability of LLMs in various applications, providing new insights into self-optimization and adaptive learning in advanced language models. The automation of role generation could revolutionize the way reasoning capabilities are designed and implemented in LLMs, contributing to the knowledge of the best Zero-Shot Reasoning techniques.

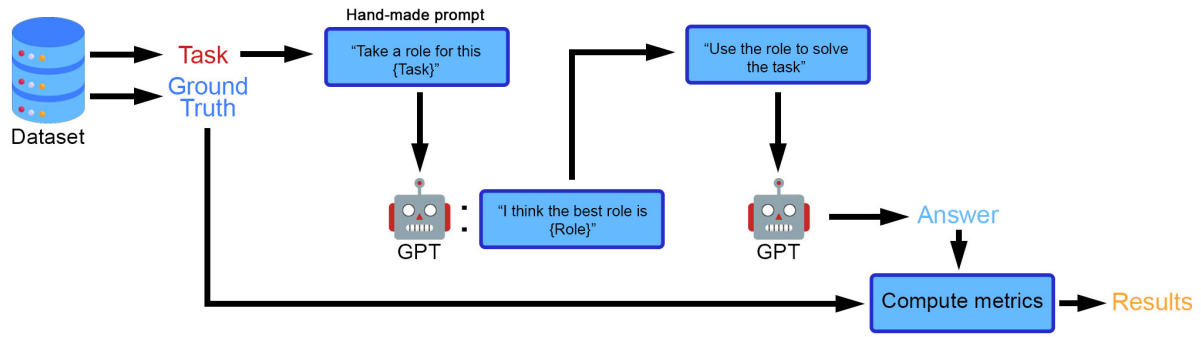

Figure 1: This block diagram illustrates the workflow of the proposed method. The dataset initially provides a task and corresponding ground truth. A hand-crafted prompt is then employed to present the task to a GPT model, which suggests an appropriate role. Subsequently, a second prompt is used to instruct the same GPT model to apply the suggested role to solve the task, thereby generating a response. Finally, metrics are computed by comparing the generated response with the ground truth, enabling an evaluation of the response's quality. It should be noted that the prompts included in this figure are not the final chosen prompts; they are used here for illustrative purposes.

2 Related Work

2.1 LLM Reasoning

Initially, large language models (LLMs) were considered to have poor reasoning abilities (Rae et al., 2022). However, the introduction of the "chain-of-thought prompting" technique by (Wei et al., 2023) marked a significant improvement in the reasoning capabilities of LLMs. This technique involves providing examples with explicit reasoning steps, which helps the models follow a logical sequence of thoughts. In the realm of Zero-Shot Reasoning, the "Zero-Shot-CoT" approach proposed by (Kojima et al., 2023) further enhances this capability. By simply adding the phrase "Let's think step by step" to the prompt, the models are encouraged to adopt a chain-of-thought reasoning process, even without prior examples.

2.2 Role-Playing

Large language models (LLMs) have demonstrated the ability to assume various roles, a feature leveraged to enhance user interaction and circumvent restrictions. Earlier works such as (Shanahan et al., 2023) and (Wu et al., 2023) explored the potential of role-playing in LLMs, examining how adopting different personas could impact performance and user engagement. The pioneering application of role-playing for reasoning tasks was presented by (Kong et al., 2024), who demonstrated that assigning specific roles to prompts could significantly improve the reasoning performance of LLMs.

In their study, (Kong et al., 2024) highlighted a critical limitation: the manual creation of prompts to describe the roles that the LLMs should follow. This manual process is time-consuming and requires a deep understanding of both the task and the model's capabilities. By addressing this challenge, our work aims to automate the generation of these role descriptions, leveraging the LLMs themselves to identify and adopt the most suitable roles for each task. This automation not only streamlines the process but also opens new avenues for optimizing and adapting LLMs to a broader range of applications.

3 Method: Adaptive Role-Playing

To better illustrate the methodology of `Adaptive Role Selection`, a graphical abstract is presented in Figure 1, which explains the proposed procedure. The core idea involves prompting the LLM to select an appropriate role based on the given task without actually solving it yet. In response to this initial prompt, the LLM provides a detailed description of the role and confirms its adoption. Subsequently, a new user input is added to the conversation, presenting the model with the task to be solved, along with an instruction to use the chosen role for solving it. Finally, the language model delivers the final response, solving the task based on the adopted role. An example of a template conversation is shown in Figure 2.

To break down the process step by step:

1. **Role Selection Prompt:** The initial prompt instructs the language model to select a suitable assistant role for the task at hand. The

Adaptive Role-Play

Input:

User: Choose the best role for the following question or task: [TASK] You should answer with a detailed description of the role you will take, not with the answer to the question or task itself.".

Output:

Assistant: In base of the task that you provided me, I think the best role is...[DETAILED DESCRIPTION OF ROLE]. Now I am an excelent....[ROLE]... and I will take that role the best I can.

Input:

User: Now using that role, answer/complete the following question/task: [TASK]

Output:

Assistant: [ANSWER]

Figure 2: Illustration of the conversation between the user and the assistant for the Adaptive Role-Play methodology

model is not required to solve the task at this stage, but merely to choose a role that it believes will be most effective.

2. **Role Description Response:** Upon receiving the initial prompt, the model responds by describing in detail the role it will assume, explaining the characteristics and approach it will take to address the task.

3. **Task Resolution Prompt:** With the role defined, a follow-up prompt is issued. This prompt presents the actual task to the model, explicitly instructing it to use the previously selected role to provide a solution.

4. **Final Response:** The model then processes the task within the context of the chosen role and generates a response, aiming to leverage the advantages of role-playing to improve its reasoning and solution quality.

4 Experiments

4.1 Tasks and Datasets

To evaluate our approach, we selected a diverse set of datasets spanning multiple reasoning categories, in line with previous research on the reasoning capabilities of LLMs (Kojima et al., 2023), (Wei et al., 2023). Specifically, we focused on two main categories: arithmetic reasoning and commonsense reasoning. The chosen datasets for each category are as follows:

1. **Arithmetic Reasoning**: - **AQUA-RAT** (Ling et al., 2017): The AQUA-RAT dataset is designed to test arithmetic reasoning and problem-solving skills. It includes complex word problems that require multi-step reasoning and calculation. Each question is presented as a multiple-choice problem, challenging the model to understand and perform the necessary calculations to arrive at the correct answer.

2. **Commonsense Reasoning**: - **CSQA** (Talmor et al., 2019): CommonsenseQA (CSQA) is a dataset focused on assessing the model's ability to apply commonsense knowledge to answer questions. The questions are derived from concepts that require an understanding of everyday human experiences and the relationships between various entities.

- **StrategyQA** (Geva et al., 2021): StrategyQA is a dataset that presents questions requiring strategic thinking and planning. Unlike typical QA datasets, StrategyQA includes questions where the answer is not directly stated but must be inferred through logical reasoning and understanding the broader context, which usually involves domain-specific information such as history, biology, geography, etc.

These datasets are selected for their ability to test the robustness and generalization capabilities of LLMs across different reasoning tasks. A notable feature of our approach is the use of a single, consistent prompt for all datasets, as presented in Figure 2. This uniformity demonstrates the adaptability and ease of implementation of our method, as it does not rely on dataset-specific prompts, thereby simplifying the deployment process.

Due to computational constraints and limited access to the model's API, we utilized 50% of the test dataset for StrategyQA and CSQA evaluations. This subset was chosen to ensure a representative sample while managing resource limitations.

4.2 Experimental Setup

For all experiments, the `gpt-3.5-turbo` model was utilized, and the approach was compared against several baselines: Zero-Shot, Zero-Shot-CoT (Chain of Thought), Few-Shot-CoT, and Role-Play Prompting. To ensure the model's responses were as deterministic as possible, the model's temperature was set to 0, thereby minimizing the influence of randomness in the generated answers.

The primary method involved providing a specific instruction to the LLM to select an appropriate assistant role for solving a given task. The LLM then used this generated role to formulate a response to the original question. After computing the responses, they were compared with the ground truth by calculating the accuracy. Subsequently, the accuracy was evaluated across the datasets of `gpt-3.5-turbo` using prompts for Zero-Shot, Zero-Shot-CoT, Few-Shot-CoT methods, and Role-Play Prompting. These results are detailed in (Kong et al., 2024).

By systematically comparing these methods, the aim was to assess the effectiveness of the adaptive role-play prompting technique. The consistent use of the same prompt across different datasets highlighted the potential of this approach to generalize well without the need for dataset-specific adjustments.

5 Results

Table 1 presents the accuracy results for each prompting methodology across the different datasets, and Table 2 shows examples of the roles chosen and responses generated by the model. For the Arithmetic dataset (Ling et al., 2017), the Role-Play Prompting methodology remains the best performing among all. However, the proposed Adaptive Role-Play methodology surpasses all other baselines, which is a significant result considering it does not require a specific prompt to be developed for each dataset. Additionally, Figure 2 includes examples of responses for the Arithmetic dataset, where it can be seen that the roles chosen do not vary much between questions since the role is generally that of a math expert. Therefore, in this case, giving the model the freedom to choose the role does not contribute significantly to performance improvements.

For the commonsense reasoning dataset CSQA (Talmor et al., 2019), an accuracy result of 74.0 is obtained, which is similar to that of Zero Shot, but does not surpass Role-Play Prompting. This is believed to be because the type of questions used in this case do not require adopting the role of an expert in a specific area, but rather require general world knowledge and common sense to answer, such as spatial and temporal sense, etc., as seen in the examples in Figure 2.

On the other hand. For the other commonsense reasoning dataset StrategyQA (Geva et al., 2021), the Adaptive Role-Play methodology outperforms all other methodologies, achieving an accuracy of 70.4%. This result demonstrates the superiority of the Adaptive Role-Play methodology in questions where some specific knowledge is needed. This is likely due to the nature of these questions, where the role that the model needs to adopt varies significantly depending on the question. The wide variety of roles taken is shown in the diverse examples of outputs displayed in Figure 2. By allowing the model the freedom to first interpret which role is most suitable for the given question and then respond in that role, improved performance is observed.

Overall, these findings highlight the potential of the Adaptive Role-Play methodology to enhance the reasoning capabilities of LLMs in various contexts, particularly in commonsense reasoning tasks where the flexibility in role selection proves to be beneficial.

6 Conclusion

This paper presents a novel approach to improving the zero-shot reasoning capabilities of large language models (LLMs) through automated role generation. By leveraging the models themselves to select the most suitable roles for each task, we have demonstrated that this method not only optimizes time and resources but also enhances performance across various reasoning tasks. The results of the experiments indicate that the Adaptive Role-Play methodology significantly improves the accuracy of LLMs, particularly in commonsense reasoning tasks, where role selection is crucial. This work contributes to the evaluation of LLMs' capabilities by developing and applying a methodology that assesses their performance in diverse reasoning contexts. Our findings underscore the potential for LLMs to autonomously generate roles, thereby reducing the need for manual prompt engineering and enhancing the scalability and adaptability of these models. This advancement opens new av-

Method	Arithmetic	Common Sense	
	AQuA	CSQA	Strategy
Zero-Shot	53.5	74.5	66.0
Zero-Shot-CoT	53.9	68.8	65.8
Few-Shot-CoT	59.4	76.3	67.4
Role-Play Prompting	**63.8**	**77.2**	67.0
Adaptive Role-Play	59.5	74.0*	**70.4***

Table 1: Accuracy comparison of Adaptive Role-Play with Role-Play Prompting, Few-Shot-CoT, Zero-Shot, Zero-Shot-CoT on each dataset. The metrics marked with * indicate that 50% of the testing dataset was used to obtain the result.

enues for self-optimization and adaptive learning in advanced language models, providing valuable insights into the future development and application of LLMs in various domains

References

Mor Geva, Daniel Khashabi, Elad Segal, Tushar Khot, Dan Roth, and Jonathan Berant. 2021. Did Aristotle Use a Laptop? A Question Answering Benchmark with Implicit Reasoning Strategies. *Transactions of the Association for Computational Linguistics*, 9:346–361.

Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2023. Large language models are zero-shot reasoners. *Preprint*, arXiv:2205.11916.

Aobo Kong, Shiwan Zhao, Hao Chen, Qicheng Li, Yong Qin, Ruiqi Sun, Xin Zhou, Enzhi Wang, and Xiaohang Dong. 2024. Better zero-shot reasoning with role-play prompting. *Preprint*, arXiv:2308.07702.

Wang Ling, Dani Yogatama, Chris Dyer, and Phil Blunsom. 2017. Program induction by rationale generation: Learning to solve and explain algebraic word problems. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 158–167, Vancouver, Canada. Association for Computational Linguistics.

Jack W. Rae, Sebastian Borgeaud, Trevor Cai, Katie Millican, Jordan Hoffmann, Francis Song, John Aslanides, Sarah Henderson, Roman Ring, Susannah Young, Eliza Rutherford, Tom Hennigan, Jacob Menick, Albin Cassirer, Richard Powell, George van den Driessche, Lisa Anne Hendricks, Maribeth Rauh, Po-Sen Huang, Amelia Glaese, Johannes Welbl, Sumanth Dathathri, Saffron Huang, Jonathan Uesato, John Mellor, Irina Higgins, Antonia Creswell, Nat McAleese, Amy Wu, Erich Elsen, Siddhant Jayakumar, Elena Buchatskaya, David Budden, Esme Sutherland, Karen Simonyan, Michela Paganini, Laurent Sifre, Lena Martens, Xiang Lorraine Li, Adhiguna Kuncoro, Aida Nematzadeh, Elena Gribovskaya, Domenic Donato, Angeliki Lazaridou, Arthur Mensch, Jean-Baptiste Lespiau, Maria Tsimpoukelli, Nikolai Grigorev, Doug Fritz, Thibault Sottiaux, Mantas Pajarskas, Toby Pohlen, Zhitao Gong, Daniel Toyama, Cyprien de Masson d'Autume, Yujia Li, Tayfun Terzi, Vladimir Mikulik, Igor Babuschkin, Aidan Clark, Diego de Las Casas, Aurelia Guy, Chris Jones, James Bradbury, Matthew Johnson, Blake Hechtman, Laura Weidinger, Iason Gabriel, William Isaac, Ed Lockhart, Simon Osindero, Laura Rimell, Chris Dyer, Oriol Vinyals, Kareem Ayoub, Jeff Stanway, Lorrayne Bennett, Demis Hassabis, Koray Kavukcuoglu, and Geoffrey Irving. 2022. Scaling language models: Methods, analysis insights from training gopher. *Preprint*, arXiv:2112.11446.

Murray Shanahan, Kyle McDonell, and Laria Reynolds. 2023. Role-play with large language models. *Preprint*, arXiv:2305.16367.

Alon Talmor, Jonathan Herzig, Nicholas Lourie, and Jonathan Berant. 2019. CommonsenseQA: A question answering challenge targeting commonsense knowledge. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4149–4158, Minneapolis, Minnesota. Association for Computational Linguistics.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2023. Chain-of-thought prompting elicits reasoning in large language models. *Preprint*, arXiv:2201.11903.

Ning Wu, Ming Gong, Linjun Shou, Shining Liang, and Daxin Jiang. 2023. Large language models are diverse role-players for summarization evaluation. *Preprint*, arXiv:2303.15078.

Examples	Task and Auto rol selecting
	AQUA-RAT
Example 1	**Question:** There are more than 501 students in a school such that 20% of them exactly took physics and 28% of them exactly took math. What could be the least possible no of students in the school? Answer Choices: (A) 550 (B) 570 (C) 600 (D) 700 (E) none of these **Auto selected rol:** I will take on the role of a mathematician or statistician to analyze the given information and determine the least possible number of students in the school. By calculating the percentages of students who took physics and math, I can use this information to find the minimum number of students that would satisfy these conditions. I will use my knowledge of percentages and proportions to arrive at the correct answer.
Example 2	**Question:** Q is as much younger than R as he is older than T. If the sum of the ages of R and T is 50 years. What is definitely the difference between R and Q's age? Answer Choices: (A) 22 (B) 27 (C) 29 (D) Cannot be determined (E) None of the above **Auto selected rol:** I will take on the role of a problem solver or mathematician for this question. As a problem solver, I will carefully analyze the information provided in the question and use logical reasoning and mathematical calculations to determine the difference between R and Q's age. I will apply algebraic techniques to set up equations based on the given information and solve for the unknown variables. By carefully following the steps and applying mathematical principles, I will be able to arrive at the correct answer to the question.
	CSQA
Example 1	**Question:** If there is a pond with trees around it, where it it likely located? Answer Choices: (A) ground (B) bathroom (C) forest (D) countryside (E) rural area **Auto selected rol:** I would take on the role of a geographer or environmental scientist to answer this question. As a geographer or environmental scientist, I would have knowledge of different types of landscapes and ecosystems. I would be able to analyze the characteristics of a pond with trees around it and determine where it is likely located based on those characteristics. By considering factors such as the presence of trees, bodies of water, and surrounding land use, I would be able to make an educated guess as to whether the pond is located in a forest, countryside, or rural area.
Example 2	**Question:** His house was a mess, he began doing housework to get what? Answer Choices: (A) boredom (B) nice home (C) michigan (D) feeling satisfied (E) house clean **Auto selected rol:** The best role for this question would be a psychologist or therapist. This role would involve understanding the underlying motivations and emotions driving the individual to clean his house. By delving into his thoughts and feelings, a psychologist or therapist can help uncover any deeper issues such as anxiety, stress, or a desire for control that may be influencing his behavior. This role would focus on providing support, guidance, and strategies for managing emotions and maintaining a clean and organized living space.
	Strategy QA
Example 1	**Question:** Did mercenaries fight for England in the Glorious Revolution? **Auto selected rol:** I would take the rol of a Historian: As a historian, I will thoroughly research and analyze primary sources, such as letters, diaries, and official documents from the time period of the Glorious Revolution. I will examine the role of mercenaries in the conflict, including any records of foreign soldiers being hired to fight for England during this time. By delving into the historical context and examining multiple perspectives, I will provide a well-informed and evidence-based answer to the question at hand.
Example 2	**Question:** Do giraffes require special facilities at zoos? **Auto selected rol:** I would take the rol of a Zoologist: A zoologist would be the best role to answer this question as they specialize in the study of animals and their behavior. They would have the knowledge and expertise to understand the specific needs and requirements of giraffes in captivity, including their habitat, diet, social interactions, and overall well-being. A zoologist would be able to provide detailed information on the type of facilities and environment that would be necessary to ensure the health and happiness of giraffes in a zoo setting.
Example 3	**Question:** Would JPEG be a good format for saving an image of Da Vinci's Vitruvian Man? **Auto selected rol:** I would take on the role of a digital imaging specialist or a graphic designer for this question. As a digital imaging specialist or graphic designer, I would have the expertise and knowledge to determine the best format for saving an image of Da Vinci's Vitruvian Man. I would consider factors such as the complexity and detail of the image, the intended use of the image, and the desired quality and resolution. By carefully analyzing these factors, I would be able to make an informed decision on whether JPEG is a suitable format for saving the image or if another format would be more appropriate.

Table 2: Examples of adaptative rol selecting by task and Dataset